

Nurture Course COURSE

PHASE : SPEED & ACCURACY TEST

TARGET : PRE-MEDICAL 2023

Test Type : **PRACTICE TEST**

Test Pattern : **NEET (UG)**

TEST DATE : _____

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	3	1	4	1	1	1	1	1	3	2	3	2	1	2	4	4	3	1	2	4	4	3	3	4	4	2	4	3	3
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45															
A.	3	3	1	2	2	3	3	2	1	1	4	3	2	4	2															



Click Here To Join our
Telegram Channel

HINT - SHEET

1. **Ans (3)**

For aluminium and copper magnitude of shear modulus is less than young's modulus and bulk modulus is maximum.

2. **Ans (3)**

From capillary tube experiment, we know that,

$$h = \frac{2S \cos \theta}{r \rho g} \quad \text{i.e. } h \propto \frac{1}{r} = \frac{h'}{h} = \frac{r}{\frac{r}{2}} = 2, \text{ or}$$

$$h' = 2h$$

3. **Ans (1)**

Per unit volume energy stored

$$= \frac{1}{2} \times Y \times (\text{strain})^2$$

$$= \frac{1}{2} \times Y \times \left(\frac{1}{L} \right)^2$$

$$\text{Given, } \frac{1}{L} \times 100 = 1\%$$

Thus per unit volume energy stored

$$= \frac{1}{2} \times 2 \times 10^{10} \times \left(\frac{1}{100} \right)^2 = 10^6 \text{ Jm}^{-3}$$

4.

Ans (4)

From the definition of bulk modulus

$$\beta = \frac{\Delta P}{\Delta V / V}$$

$$\frac{-(1.65 - 1.01)10^5}{-10} \times 100 = 1.55 \times 10^5$$

5.

Ans (1)

$$\text{Terminal velocity, } v = \frac{2r^2 (\rho - \rho_0) g}{9}$$

$$\text{i.e. } v \propto r^2$$

$$\therefore \frac{v_1}{v} = \frac{r_1^2}{r^2}$$

$$\text{Or, } v_1 = v \frac{r_1^2}{r^2} = 20 \left(\frac{1}{2} \right)^2 = 5 \text{ cms}^{-1}$$

6. **Ans (1)**

$$\text{Poisson's ratio } (\sigma) = \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

$$0.4 = \frac{10^{-5}}{\frac{\Delta L}{L}}$$

$$\frac{\Delta L}{L} = \text{longitudinal strain} = 2.5 \times 10^{-5}$$

Young's Modulus

$$Y = \frac{\text{Stress}}{\text{Strain}}$$

$$= \frac{F/A}{\Delta L/L}$$

$$= \frac{F}{A(\Delta L/L)}$$

$$= \frac{100}{0.025 \times 10^{-4} \times 2.5 \times 10^{-5}}$$

$$= 1.6 \times 10^8 \text{ N m}^{-2}$$

7. **Ans (1)**

$$\text{Energy Stored} = U = \frac{1}{2} Y \times (\text{strain})^2 \times \text{volume}$$

$$\text{or } U = \frac{1}{2} Y \times \left(\frac{x}{l}\right)^2 \times Al$$

$$\text{or } U = \frac{1}{2} \frac{Yx^2}{l} \times A$$

$$\text{or } U = \frac{1}{2} \frac{YA}{l} x^2$$

8. **Ans (1)**

Let the atmospheric pressure be P_0 pressure just outside the bubble,

$$P = P_0 + h\rho g$$

Pressure of air inside the bubble

$$P_{\text{in}} = P + \frac{2T}{r}; T \text{ is surface tension}$$

$$P_{\text{in}} = P_0 + h\rho g + \frac{2T}{r}$$

$$\therefore P' - P_0 = \frac{1}{5} \times 1000 \times 9.8 + \frac{2 \times 0.075}{1 \times 10^{-3}}$$

$$= 1960 + 150 = 2110 \text{ Pa}$$

9. **Ans (1)**

$$\frac{2T}{r} = h\rho g$$

10. **Ans (3)**

$$t \propto \sqrt{H}$$

12. **Ans (3)**

Volume flowing per second in the left tube =
(Volume flowing per second 1st branch tube) +
(volume flowing per second in 2nd branch tube)

$$Av_1 = Av_2 + 1.5Av$$

$$A \times 3 = A \times 1.5 + 1.5Av$$

$$v = 1 \text{ m/s}$$

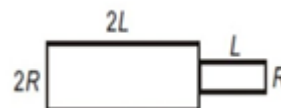
13. **Ans (2)**

$$3P = P + \rho gh$$

$$P = P + \rho g \left(h - \frac{h}{5} \right) = P + \frac{4}{5} \rho gh$$

$$= P + \frac{4}{5} 2P = \frac{13}{5} P$$

14. **Ans (1)**



$$\Delta \ell = \frac{FL}{AY}$$

$$\frac{\Delta \ell_1}{\Delta \ell_2} = \frac{L_1}{A_1} \times \frac{A_2}{L_2} = \frac{2L}{\pi(2R)^2} \times \frac{\pi R^2}{L} = \frac{1}{2}$$

15. **Ans (2)**

$$\Delta l = \frac{(\text{stress}) (l)}{Y}$$

$$= \frac{10^9}{2 \times 10^{11}} \times 2.8 \text{ (in } \text{\AA})$$

$$= 1.4 \times 10^{-2} \text{ \AA}$$

$$= 14 \times 10^{-3} \text{ \AA}$$

16. **Ans (4)**

For the blood flow in capillary, volumetric flow rate = velocity $\times$ area of cross section

$$V = \frac{Q}{A} = \frac{100 \times 10^{-6}}{0.25}$$

$$V = 400 \times 10^{-3} \text{ mm/s}$$

17. Ans (4)

There will be two surface formed (front side and back side) and force due to surface tension will be given by :

$$F = TL + TL = 2TL$$

Force must balance the weight to keep it in equilibrium

$$W = 2TL$$

$$mg = 2TL$$

$$T = \frac{mg}{2L} = \frac{1.5 \times 10^{-2}}{2 \times 0.3} = 0.025 \text{ Nm}^{-1}$$

18. Ans (3)

For equilibrium, the total up thrust will be equal to the weight of the sphere. If V is the volume of the sphere, we have

$$\left(\frac{V}{2}\right)(13.6)g + \left(\frac{V}{2}\right)(0.8)g = V\rho g$$

$$\rho = \left(\frac{13.6 + 0.8}{2}\right) \text{ gcm}^{-3} = 7.2 \text{ gcm}^{-3}$$

19. Ans (1)

Viscosity of Glycerin is greatest among the three liquids and therefore large viscous forces brings it to rest.

20. Ans (2)

Difference of pressure between sea level and the top of hill

$$\Delta P = (h_1 - h_2) \times \rho_{\text{Hg}} \times g$$

$$= (75 - 50) \times 10^{-2} \times \rho_{\text{Hg}} \times g \dots (i)$$

And the pressure difference due to h meter of air

$$\Delta P = h \times \rho_{\text{air}} \times g \dots (ii)$$

By equating (i) and (ii) we get

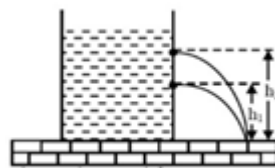
$$h \times \rho_{\text{air}} \times g = (75 - 50) \times 10^{-2} \times \rho_{\text{Hg}} \times g$$

$$h = 25 \times 10^{-2} \left(\frac{\rho_{\text{Hg}}}{\rho_{\text{air}}}\right) = 25 \times 10^{-2} \times 10^4$$

$$= 2500 \text{ m}$$

so, height of the hill = 2.5 km

21. Ans (4)



If H is the height of the liquid surface, then for same range

$$H = h_1 + h_2$$

And for maximum range

$$h = \frac{H}{2} = \frac{h_1 + h_2}{2}$$

22. Ans (4)

This happens in accordance with equation of continuity and this equation was derived on the principle of conservation of mass and it is true in every case, either tube remain horizontal or vertical.

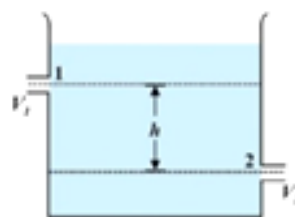
23. Ans (3)

According to Pascal's Law

$$\frac{f_1}{f_2} = \frac{A_1}{A_2} = \frac{r_1^2}{r_2^2} = \frac{1}{4}$$

$$f_1 = \frac{1}{4} Mg$$

24. Ans (3)



Rate of change of momentum = dp/dt

$$F = F_2 - F_1 = A_2 p_2 \times v_2 - A_1 p_1 \times v_1$$

$$= A_2 p_2 (v_2^2 - v_1^2) \dots \dots \dots (i)$$

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho gh = P_2 + \frac{1}{2} \rho v_2^2 + 0$$

$$(\because P_1 = P_2)$$

$$\Rightarrow \frac{1}{2} \rho (v_2^2 - v_1^2) = \rho gh$$

$$\therefore F = 2A_2 \rho gh$$

25. **Ans (4)**

After travelling certain distance drop will attain maximum velocity called terminal velocity and this will remain constant for constant medium and independent of h .

26. **Ans (4)**

As cross-section areas of both the tubes A and C are same and tube is horizontal. Hence according to equation of continuity

$$v_A = v_C$$

Therefore, according to Bernoulli's theorem

$$P_A = P_C$$

i.e., height of liquid is same in both the tubes A and C.

27. **Ans (2)**

If two liquids of equal masses and different densities are mixed together then the density of the mixture is

$$\rho = \frac{2\rho_1\rho_2}{\rho_1 + \rho_2} = \frac{2 \times 1 \times 2}{1 + 2} = \frac{4}{3} \text{ g/cc}$$

28. **Ans (4)**

The length of the circular plate on which the force will act = $2\pi R$

Force to pull = (surface tension) $\times$ (Length)

$$F = 75 \times 2\pi \times 5 = 750\pi \text{ dynes}$$

29. **Ans (3)**

Increase in length is given by

$$l = \frac{FL}{AY}$$

$$\therefore l \propto \frac{1}{r^2} \text{ (F and Y are constant)}$$

$$\frac{l_2}{l_1} = \left(\frac{r_1}{r_2}\right)^2 = 2 = 4$$

$$\therefore l_2 = 4l_1 = 4\text{cm}$$

30. **Ans (3)**

Viscous force is given by

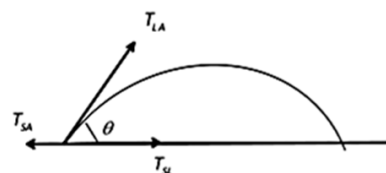
$$F = \eta A \frac{v}{d}$$

$$\frac{F_1}{F_2} = \frac{V_1}{V_2}$$

$$\frac{800}{2400} = \frac{2}{V_2}$$

$$v_2 = 6\text{cm/s}$$

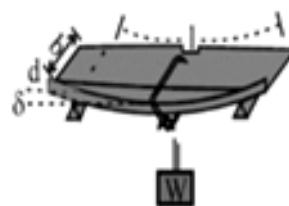
31. **Ans (3)**



$$\cos\theta = \frac{T_{SA} - T_{SL}}{T_{LA}}$$

From this diagram, angle of contact depends on surface tension due to solid, liquid and gas(air) all the three

32. **Ans (3)**



δ = Depression in the beam

Y = Young's Modulus

W = Load applied

L = Length of the beam

b = width of the beam

d = thickness of the beam

$$\delta = \frac{WL^3}{4Ybd^3}$$

$$\therefore \delta \propto \frac{1}{Y}$$

33. **Ans (1)**

In Bernoulli's theorem

$$\text{Velocity head} = \frac{v^2}{2g}$$

$$\text{Velocity head} = \frac{(5)^2}{2 \times 10}$$

$$\text{Velocity head} = 1.25 \text{ m}$$

34. **Ans (2)**

Terminal speed is

$$v = \frac{2r^2g}{9\eta}(\sigma - \rho)$$

Therefore $v \propto r^2$

$$\frac{r_1}{r_2} = \sqrt{\frac{v_1}{v_2}}$$

$$\frac{r_1}{r_2} = \sqrt{\frac{4}{9}}$$

$$\frac{r_1}{r_2} = \frac{2}{3}$$

35. **Ans (2)**

Bernoulli's theorem for unit mass of liquid is

$$\frac{P}{\rho} + \frac{1}{2}v^2 = \text{constant}$$

As the liquid starts flowing, its pressure energy

decreases

$$\begin{aligned} \frac{1}{2}v^2 &= \frac{\Delta P}{\rho} \\ \Rightarrow \frac{1}{2}v^2 &= \frac{3.5 \times 10^5 - 3 \times 10^5}{10^3} \\ \Rightarrow v^2 &= \frac{2 \times 0.5 \times 10^5}{10^3} \\ \Rightarrow v^2 &= 100 \\ \Rightarrow v &= 10 \text{ m/s} \end{aligned}$$

36. **Ans (3)**

According to Hooke's law, within the elastic limits stress is directly proportional to strain.

$$\text{or, } \frac{\text{stress}}{\text{strain}} = \text{constant} = E$$

The constant E is called the modulus of elasticity.

37. **Ans (3)**

A liquid surface comes at rest after stirring due to viscosity. Rest are examples of surface tension.

38. **Ans (2)**

Young's modulus of wire does not vary with the dimensions of the wire. It is a property of the material.

39. **Ans (1)**

From Pascal's law, if pressure is increased at one point then it gets increased at all points in the entire liquid by the same amount. Therefore, if an additional force F is applied at each piston, then pressure at each point and therefore also at the centre of the container increases by the same amount of $\frac{F}{A}$.

40. **Ans (1)**

$$\text{Because, } h = \frac{2S}{\rho g} \Rightarrow S = \frac{\rho g h}{2}$$

$$W_{ST} = 2\pi r S h = \frac{4\pi S^2}{\rho g}$$

$$\text{and } W_g = \frac{(\pi r^2 h \rho) g h}{2} = \frac{2\pi S^2}{\rho g}$$

$$\text{Heat} = Q = W_{ST} - W_g$$

$$Q = \frac{2\pi S^2}{\rho g}$$

41. **Ans (4)**

The terminal velocity is given by

$$v = \frac{2r^2(\rho - \rho_o)g}{9\eta}$$

$$\text{or } \frac{v}{r^2} = \frac{2(\rho - \rho_o)g}{9\eta} = \text{constant}$$

42. **Ans (3)**

Gravitational force remains constant on the falling spherical ball. It is represented by straight line P.

The viscous force $F = 6\pi\eta r v$ increases as the velocity increases with time. Hence, it is represented by curve Q.

Net force = gravitational force — viscous force
As viscous force increases, net force decreases and finally becomes zero. Then the body falls with a constant terminal velocity. It is thus represented by curve R.

43. **Ans (2)**

Excess pressure inside a bubble just below the surface of water is

$$P_1 = \frac{2T}{r}$$

and pressure inside the drop just outside the surface is

$$P_2 = \frac{2T}{r}$$

Hence $P_1 = P_2$

44. **Ans (4)**

When a tiny lead shot is gently dropped on the surface of a viscous liquid, the velocity of the lead shot increases with time until it finally attains a steady value known as terminal velocity.

45. **Ans (2)**

The height up to which water will rise

$$l = \frac{h}{\cos \alpha} = \frac{2\text{cm}}{\cos 60^\circ} = 4.0\text{cm}$$

h = vertical height

α = Angle with vertical